APMTH 50 Problem Set 5

Due on Friday, April 21, by 5pm

Collaboration policy

Collaboration on homework and labs is encouraged, but each student must write solutions in their own words, and not copy them from anyone else. Please write the name(s) of your collaborator(s) at the top of the homework.

Submission guidelines

Please upload your homework as a single Word or pdf document. **For this homework, you should submit all of your code as well as the output from your code.**

Final project workshops

Classes this week will be final project workshops. Please come prepared to discuss your project and what you have done on it so far. See comments on your homework 4 Canvas submission for ideas on how to prepare for these workshops.

Problems

1. **Unique number strategy.** Submit your Matlab function for the unique number competition, following the instructions and information from the lab. Your routine should be called `firstname_lastname.m` and should be uploaded as a separate file to Canvas.

2. **Rock-paper-scissors.** In this well-known game, two players simultaneously choose between the options of rock, paper, or scissors. The winner is then decided by using the following conditions: rock beats scissors, scissors beats paper, and paper beats rock.

   (a) Suppose all that matters is the total number of wins, so that the payoff matrix for the first player is

   $$A = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}. \tag{1}$$

   Consider playing a mixed strategy where rock, paper, and scissors are chosen with probabilities p_0, p_1, and p_2 respectively. Calculate the expected payoff for playing either rock, paper, or scissors. Use these to determine a set of three differential equations for p_0, p_1, and p_2 for replicator dynamics following the procedure described in the lab. Starting from $(p_0, p_1, p_2) = (0.8, 0.1, 0.1)$, solve the system of differential equations numerically to find equilibrium values of p_0, p_1, and p_2.

(b) Now suppose that a win scores one point, and a loss loses one point, so that the payoff matrix for the first player is

$$A = \begin{pmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{pmatrix}. \tag{2}$$

Modify your program from part (a) to solve the replicator dynamics for this payoff matrix. Try the following two initial conditions: (i) your equilibrium solution from part (a), and (ii) $(p_0, p_1, p_2) = (0.8, 0.1, 0.1)$. Comment on the results.

(c) **Extra credit.** Examine the case where a win scores one point and a loss loses two points.

3. **Replication dynamics for a small unique number game.** Consider a unique number game where three players A, B, and C choose from the numbers 0, 1, and 2.

   (a) Suppose that A chooses 0. What are the possible choices of the other two players that will lead to A winning? Repeat this for when A chooses 1, and when A chooses 2.

   (b) Suppose that the three players are employing a mixed strategy in which the numbers 0, 1, and 2 are chosen with probabilities p, q, and r respectively. Suppose that winning a round scores one, and that all other outcomes score zero. By making use of your results from part (a), calculate the expected payoff of choosing 0, 1, or 2.

   (c) Use your results from part (b) to set up a replicator dynamics system of three differential equations for p, q, and r following the procedure described in the lab. Solve the system and determine an equilibrium solution, making sure that you simulate the system over a long enough time interval to obtain three digits of precision in the equilibrium probabilities.

   (d) **Extra credit.** Repeat the analysis in steps (a), (b), and (c) for the following extensions to the above scenario:

      i. Four players choosing from 0, 1, and 2.

      ii. Three players choosing from 0, 1, 2, and 3.

      iii. Three players choosing from 0, 1, and 2, where a player scores $\frac{1}{3}$ when no unique number is chosen.

   (e) **Extra credit.** Suppose that you are playing many repeated games, and your objective is to win more games than your opponents. What is a better strategy: your answer from (c) or your answer from (d)(iii)?